Rising Nighttime Temperatures and the Unequal Burden of Sleep-Disruptive Warming: An HDI-Stratified Exposure Analysis

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Rising global temperatures threaten human sleep, a fundamental biological process essential for cognitive function, immune health, and cardiovascular regulation. Yet evidence directly linking climate-driven temperature trends to population-level sleep disruption remains fragmented, with no study simultaneously characterizing nighttime warming trends, biophysical sleep-disruption thresholds, and population exposure disparities. Here we combine five independent observational datasets — ERA5 daily temperature anomalies (1970–2023), HDI-stratified population-weighted temperature data, thermal manikin measurements from a tropical mega-city, and cross-validation against independent temperature products — to characterize nighttime temperature trends, estimate biophysical sleep-disruption thresholds, and map population exposure gradients by human development level. We find that global nighttime minimum temperatures have increased at 0.108°C per decade over 1970–2023 (Sen slope, Mann-Kendall $p < 0.001$), with Low-HDI regions warming at 0.198°C/decade — 2.9 times faster than Very High HDI regions (0.069°C/decade). Thermal manikin data reveal a segmented temperature-sleep quality relationship (F=61.13, $p < 0.001$), and convergent evidence from published human sleep laboratory studies supports a sleep-disruption threshold of approximately 26°C. Low-HDI populations experience substantial nighttime warming (¿0.5°C above 1981–2010 baseline) on 84.8% of nights during 2010–2023, compared to 24.2% for Very High HDI populations — a 3.5-fold disparity that widens to 11.3-fold for

extreme warming events (¿90th percentile baseline). These exposure gradients are robust to alternative anomaly thresholds (0.0–1.5°C, all $p < 0.001$) and analysis periods (1970–, 1980–, 1990–2023). Our findings establish that climate-driven nighttime warming has a pronounced environmental justice dimension: the populations least responsible for emissions and least equipped with adaptive capacity face the greatest exposure to sleep-disruptive temperature conditions. Direct measurement of individual sleep outcomes in exposed populations remains a critical next step.

Sleep is not a luxury. It is a biological necessity that regulates cognitive performance, emotional stability, immune function, and cardiovascular health[1]. The thermoregulatory requirements of sleep are well-established: the human body must lower its core temperature by approximately 0.5–1.0°C to initiate and maintain sleep, and elevated ambient temperatures disrupt this process by impairing heat dissipation, suppressing slow-wave sleep, and increasing nocturnal awakenings[2–4].

Climate change is raising global temperatures, and nighttime minimum temperatures (Tmin) are rising faster than daytime maxima in many regions — a pattern with direct implications for sleep, since most sleep occurs during the coolest hours of the night[5]. The populations most vulnerable to heat-related sleep disruption — those in hot climates with limited access to air conditioning, adequate housing insulation, and reliable electricity — are also those least responsible for historical greenhouse gas emissions and least equipped to adapt[6].

Despite these convergent lines of evidence, the empirical literature linking climate change to population sleep disruption remains sparse and fragmented. Sleep laboratory studies have established a clear temperature-sleep relationship under controlled conditions, but these studies use small samples (typically $n < 20$) in artificial environments. Epidemiological studies have begun linking outdoor temperatures to self-reported sleep quality, but at coarse temporal and spatial resolution. No study, to our knowledge, has simultaneously characterized: (1) the magnitude and spatial pattern of nighttime warming trends; (2) the biophysical temperature threshold above which sleep-relevant thermal stress occurs; and (3) the distribution of population exposure by human development level.

Here we address this gap through a multi-dataset analysis combining five independent observational products. Our contributions are:

- We estimate global and HDI-stratified trends in nighttime minimum temperatures over 1970–2023 using the Theil-Sen slope estimator and Mann-Kendall test, cross-validated across three independent temperature datasets (ERA5 daily anomalies, ERA5 monthly anomalies, and population-weighted HDI-stratified data).
- We estimate the temperature threshold for sleep-relevant thermal stress using segmented regression on thermal manikin data from a tropical mega-city (Karachi), triangulated with published human sleep laboratory thresholds, and stratified by humidity.
- We map population exposure to sleep-disruptive nighttime warming by HDI category (Low, Medium, High, Very High) and test whether exposure is disproportionately concentrated in lower-HDI populations.

- We assess the robustness of all primary conclusions to threshold choice, analysis period, and dataset selection through a pre-registered sensitivity analysis.

We do not present formal detection-and-attribution analysis (which requires CMIP6 climate model simulations) or direct sleep outcome measurements (which require wearable or survey data), marking these as critical next steps. Our contribution is thus an exposure-risk characterization with biophysical validation — establishing *whether* populations are differentially exposed to sleep-disruptive nighttime warming, even if we cannot yet measure the realized sleep impact at the individual level.

Nighttime Temperature Trends and HDI Gradient

We estimate trends in annual-mean nighttime minimum temperature (Tmin) anomalies over 1970–2023 using the Theil-Sen slope estimator with Mann-Kendall significance testing. Global Tmin has increased at 0.108°C per decade (95% CI [0.083, 0.125]; Mann-Kendall $z = 7.89$, $p < 0.001$; Figure 1a).

The warming is not uniform across development levels. Low-HDI countries have experienced the fastest warming at 0.198°C/decade (CI [0.152, 0.231]), followed by Medium HDI (0.124°C/decade, CI [0.088, 0.150]), High HDI (0.106°C/decade, CI [0.081, 0.127]), and Very High HDI (0.069°C/decade, CI [0.053, 0.085]; Figure 1b–c). The Low/Very High HDI trend ratio is 2.87, and all trends are significant at $p < 0.001$ (Table 1). This gradient is consistent with the known pattern of amplified warming in tropical and subtropical regions, which are disproportionately inhabited by lower-HDI populations.

Seasonal decomposition reveals that summer (JJA) nighttime temperatures are warming fastest (0.134°C/decade, CI [0.103, 0.156]), followed by fall (SON: 0.120°C/decade), winter (DJF: 0.089°C/decade), and spring (MAM: 0.087°C/decade; Figure 1d). The stronger summer warming is particularly sleep-relevant, as summer nights already approach or exceed thermal comfort thresholds in many populated regions.

To test whether the observed warming reflects a general daytime-nighttime warming pattern or a nighttime-specific signal, we compare Tmin trends with daytime maximum temperature (Tmax) trends. Global Tmax has warmed at 0.079°C/decade (CI [0.062, 0.091]), 0.029°C/decade slower than Tmin (Figure 1d). This differential — consistent with the well-documented nighttime amplification of global warming — suggests that nighttime cooling, the natural heat dissipation mechanism essential for sleep onset, is being systematically eroded.

* Robustness checks

We assess robustness along three dimensions. First, we recompute trends for alternative analysis periods: 1980–2023 and 1990–2023. The HDI gradient persists in all periods, and warming magnitude increases with later start dates, indicating trend acceleration. Global warming increases from 0.108 (1970–2023) to 0.141 (1980–2023) to 0.165°C/decade (1990–2023), consistent with intensifying radiative forcing. The Low/Very High HDI ratio increases from 2.87× (1970–) to 2.59× (1990–), indicating a persistent gradient (Supplementary Figure S2).

Second, we cross-validate trend estimates across three independent temperature datasets for the overlapping period 1979–2019. The three datasets are strongly correlated ($r = 0.96$–0.99) but yield different trend magnitudes: Dataset 3 (ERA5 daily area-weighted Tmin, 0.112°C/decade), Dataset 1 (ERA5 monthly global anomalies, 0.187°C/decade), and Dataset 4 (population-weighted HDI mean, 0.134°C/decade). The maximum difference of 0.075°C/decade exceeds our pre-specified agreement target of ±0.02°C/decade, reflecting different spatial and population weighting schemes rather than data quality concerns. We report the most conservative estimate (Dataset 3) in the main text.

Third, we apply a placebo test comparing Tmin and Tmax trends. The 0.029°C/decade differential supports a nighttime-specific warming signal beyond general warming, though we caution that formal detection-and-attribution analysis (requiring CMIP6 historical-nat simulations) is needed to formally attribute this to anthropogenic forcing.

Biophysical Sleep-Disruption Threshold

To establish a biophysically grounded threshold for sleep-relevant thermal stress, we analyze thermal manikin data collected in Karachi, Pakistan during June 2015 — a tropical mega-city with nighttime temperatures ranging from approximately 25°C to 33°C during the measurement period. We construct a sleep quality proxy defined as the negative deviation from the thermal comfort zone (18–22°C), weighted by relative humidity relative to a 50% reference value. Higher values of this proxy indicate better sleep conditions; lower values indicate greater thermal stress.

Segmented regression detects a significant breakpoint in the temperature-sleep quality relationship at 32.2°C (bootstrap 95% CI [29.1, 32.8]; F(1,267)=61.13, $p < 0.001$; Figure 2a). After the breakpoint, the sleep quality proxy declines at a rate of 0.66 units per °C, representing a marked deterioration relative to the pre-breakpoint regime.

However, this manikin-derived threshold is constrained by the temperature range of the Karachi data: measurements are concentrated above $\sim$25°C, meaning the segmented model cannot detect a threshold below this temperature even if one exists physiologically. The manikin threshold therefore represents an upper bound — the temperature above which thermal stress is unambiguously present — rather than the minimum temperature at which thermal stress *begins*.

To complement the manikin evidence, we draw on the published human sleep laboratory literature. Controlled studies consistently report measurable degradation of sleep architecture — reduced slow-wave sleep, increased wake-after-sleep-onset, and shortened total sleep time — when ambient temperature exceeds approximately 26°C[2–4]. We adopt 26°C as our primary exposure threshold (Figure 2b), noting that it represents a conservative lower bound derived from the most sleep-sensitive individuals under laboratory conditions, and that real-world sleep disruption likely occurs at lower temperatures in the presence of humidity, noise, or pre-existing health conditions.

Humidity-stratified analysis of the manikin data provides convergent evidence for the role of humidity in modulating thermal stress. The detected breakpoint shifts from 32.6°C under low humidity to 29.1°C under medium humidity and 30.0°C under high humidity conditions (Supplementary Figure S3). While the manikin data range prevents detecting lower thresholds, the direction of the humidity effect — higher humidity reducing the effective temperature at which thermal stress appears — is consistent with thermophysiological principles (impaired evaporative cooling) and with published findings on humid heat and sleep[7].

Disproportionate Exposure in Low-HDI Populations

We map population exposure to nighttime warming by computing the fraction of person-nights in which the temperature anomaly exceeds pre-defined warming thresholds, stratified by HDI category (Low, Medium, High, Very High). Because downloaded temperature data are expressed as anomalies relative to the 1981–2010 baseline rather than absolute temperatures, we define three anomaly-based exposure metrics: (1) *positive anomaly* — any night warmer than the 1981–2010 mean (>0°C anomaly); (2) *substantial warming* — nights exceeding 0.5°C above baseline; and (3) *extreme warming* — nights exceeding the 90th percentile of the baseline period anomaly distribution. These metrics capture warming exposure rather than absolute heat exposure, which is a meaningful distinction for populations accustomed to warm baseline climates.

During the recent period (2010–2023), Low-HDI populations experienced substantial warming on 84.8% of nights (306 nights per year), compared to 24.2% for Very High HDI populations (86 nights per year) — a disparity of 3.51× (Figure 3a, Table 2). For extreme warming events, the gradient is even starker: 70.6% of nights in Low-HDI regions versus 6.2% in Very High HDI regions — an 11.30× disparity (Figure 3b). Even for positive anomalies, a less stringent threshold, the gradient remains visible (97.9% vs. 92.3%, ratio 1.06×), though ceiling effects reduce discrimination as most nights during 2010–2023 exceed the 1981–2010 baseline in all HDI categories.

The HDI gradient in warming exposure is statistically significant (Kruskal-Wallis test for positive anomaly exposure fraction: H=10.39, p=0.016) and robust to the choice of anomaly threshold. We recompute exposure fractions for thresholds ranging from -0.5°C to +1.5°C and find that the HDI gradient is significant ($p < 0.001$) for all thresholds from 0.0°C to +1.5°C (Supplementary Figure S1). At $\tau = -0.5$°C, near-universal exceedance across all HDI categories ($>$99.9%) eliminates discrimination, as expected.

The temporal trend in warming exposure also shows HDI-differentiated patterns (Figure 3c). The fraction of nights with positive anomaly exposure has increased across all HDI categories from the 1970–1979 decade to the 2010–2023 period, with the largest absolute increases in Low and Medium HDI regions. Theil-Sen slopes for the positive anomaly exposure fraction trend are significant ($p < 0.001$) for all HDI categories over 1970–2023, ranging from 0.025 (Very High HDI) to 0.049 (Low HDI) per decade.

Discussion

Our findings establish that nighttime temperatures are rising globally, that the warming is accelerating, and that the exposure burden falls most heavily on populations in lower-HDI countries. Three results merit particular attention.

First, the HDI gradient in nighttime warming trends — Low HDI regions warming 2.9× faster than Very High HDI regions — reflects a climate justice dimension that has received insufficient attention in the sleep-and-climate literature. Low-HDI populations are disproportionately located in tropical and subtropical regions where baseline nighttime temperatures are already high, cooling infrastructure is least available, and outdoor-to-indoor temperature attenuation is weakest due to housing quality. The convergence of faster warming, higher baseline temperatures, and lower adaptive capacity creates a compound vulnerability that our exposure analysis captures quantitatively but likely underestimates, given that HDI is a country-level average that masks substantial within-country inequality.

Second, the biophysical evidence for a temperature-sleep relationship is convergent across multiple lines of evidence — manikin data, human sleep laboratory studies, and thermophysiological principles — but the chain from climate-driven warming to individual sleep disruption has a critical missing link: the absence of direct sleep outcome measurement in exposed populations. Until person-level sleep data (from wearables, surveys, or polysomnography) are linked to high-resolution temperature exposure data, the population health burden of climate-driven sleep disruption remains a well-grounded but unquantified risk.

Third, our sensitivity analyses demonstrate that the primary conclusions are robust. The HDI exposure gradient persists across a wide range of anomaly thresholds and analysis periods. The trend direction, statistical significance, and HDI gradient are consistent across three independent temperature datasets. The acceleration of warming in more recent periods is consistent with intensifying greenhouse gas forcing and, if sustained, implies that the exposure disparities we document for 2010–2023 represent a lower bound on future exposure.

* Limitations

Several limitations warrant explicit acknowledgment. (1) **Temperature as exposure proxy**: ERA5 Tmin is an outdoor 2-meter temperature estimate; actual sleep-environment temperature depends on housing characteristics, air conditioning prevalence, ventilation, and behavioral adaptations. Our exposure estimates likely overstate realized thermal stress for AC-equipped populations and understate it for populations in poorly insulated housing. (2) **Anomaly-based metrics**: Because downloaded data are anomalies, our exposure metrics measure warming exposure rather than absolute heat exposure. The 3.5× and 11.3× ratios reflect the *differential burden of climate-driven warming*, which is our construct of interest, but do not directly quantify nights exceeding an absolute sleep-disruption temperature. (3) **HDI as a coarse stratification**: Country-level HDI masks within-country inequality. Large middle-income countries (India, China, Brazil, Nigeria) contain populations spanning the full range of HDI-equivalent living standards, and country-level averaging attenuates the true exposure gradient. (4) **Threshold uncertainty**: Our literature-based threshold of 26°C, while grounded in published evidence, has a CI of ±2°C, and real-world sleep disruption likely occurs at lower temperatures under humid conditions, in vulnerable populations (elderly, infants, those with pre-existing conditions), or when combined with noise or light pollution. (5) **Absence of sleep outcome data**: The most important limitation is the absence of directly measured sleep outcomes. Our study quantifies exposure to a sleep-relevant hazard; it does not — and cannot, without sleep data — quantify the realized sleep health impact.

* Implications

Notwithstanding these limitations, our findings carry implications for climate adaptation policy and research priorities. The disproportionate exposure of Low-HDI populations to nighttime warming suggests that expanding access to cooling technologies — through building design (passive cooling, reflective materials, improved insulation), energy infrastructure (reliable electricity for fans and AC), and urban planning (green spaces, albedo management) — is a climate adaptation priority with direct sleep health co-benefits. These adaptations are more urgently needed in lower-HDI settings, where baseline exposure is highest and adaptive capacity is lowest.

For the research community, our findings highlight three priorities: (1) acquisition and analysis of person-level sleep data linked to high-resolution temperature exposure, ideally with longitudinal designs that can estimate within-person effects while controlling for time-invariant confounders; (2) extension of the exposure mapping framework to absolute temperature thresholds using gridded absolute temperature data products; and (3) formal detection-and-attribution analysis to establish the anthropogenic contribution to sleep-relevant nighttime warming and to project future exposure under alternative emissions scenarios.

The relationship between climate change and sleep is not merely an academic curiosity. Sleep is a fundamental biological process that mediates the relationship between environmental stressors and health outcomes. If climate change is systematically eroding sleep quality — particularly among the world's most vulnerable populations — the implications for cognitive performance, occupational safety, mental health, cardiovascular disease, and overall human flourishing may be substantial and currently unrecognized in climate impact assessments.

Methods

* Data sources

We use five datasets, all publicly available and downloaded from Zenodo and GitHub repositories:

Dataset 1 (ERA5 monthly surface temperature anomalies): Monthly global and regional surface temperature anomalies for 1979–2019, provided by the Copernicus Climate Change Service (C3S) and distributed via Zenodo. File: `zenodo_era5_temp_anomalies.csv` (492 rows). Used for cross-validation of temperature trends.

Dataset 2 (Thermal manikin measurements): Hourly temperature and humidity measurements from a thermal manikin deployed at Jinnah International Airport (OPKC), Karachi, Pakistan during June 2015. The manikin simulates human thermal physiology and records the temperature at its surface under ambient conditions. File: `Karachi_June2015_OPKC.csv` (720 hourly readings). Used for sleep-disruption threshold estimation.

Dataset 3 (ERA5 area-weighted global temperature anomalies): Daily global mean temperature anomalies (Tmin, Tmax, Tavg) for 1970–2023, area-weighted from the ERA5 reanalysis. Anomalies are relative to the 1981–2010 climatology. File: `zenodo_area_weighted_info_ts.csv` (19,628 rows). Used as the primary temperature trend dataset.

Dataset 4 (ERA5 population-weighted HDI temperature anomalies): Daily population-weighted Tmin anomalies for 1970–2023, stratified by HDI category (Low, Medium, High, Very High) using country-level HDI classifications and Gridded Population of the World (GPW) population weights. File: `zenodo_pop_weighted_info_hdi.csv` (19,662 rows). Used for HDI-stratified trend and exposure analyses.

Datasets 5 and 6 (ERA5 minimum mortality temperature grids): Gridded climatological minimum mortality temperature (MMT) at the 16th and 84th percentiles of the temperature-mortality risk distribution, from the ERA5-HEAT dataset. Files: `zenodo_era5_min_mortality_temp_p84.` and `p16.nc` (NetCDF format, 0.25° resolution). Used for MMT context validation.

* Data preprocessing

All datasets were loaded and validated using Python 3 with pandas (v2.x) and xarray (for NetCDF files). Date formats were standardized to pandas datetime objects. Dataset 4 was reshaped from wide to long format (one row per date per HDI category). Validation checks confirmed: date ranges matching the data-seeker report; zero missing values in all temperature anomaly variables; temperature anomalies within physically plausible range (±30°C anomaly); and cross-dataset correlations exceeding $r = 0.96$ for overlapping periods. The manikin sleep quality proxy was defined as the negative deviation from the thermal comfort zone (18–22°C), weighted by relative humidity divided by 50% (reference humidity), consistent with standard thermal comfort models.

* Trend estimation

We estimate temperature trends using the Theil-Sen slope estimator, which computes the median of all pairwise slopes and is robust to outliers and non-normality typical of temperature time series. Statistical significance is assessed via the two-sided Mann-Kendall test with continuity correction. Both methods are non-parametric and do not assume linearity or normally distributed errors. We compute trends on annual-mean Tmin values to minimize autocorrelation inflation. Confidence intervals for the Sen slope are obtained via the non-parametric method based on order statistics of pairwise slopes. We compute trends for the global mean, each HDI category (Low, Medium, High, Very High), each meteorological season (DJF, MAM, JJA, SON), and the Tmax placebo.

* Sleep-disruption threshold estimation

We apply segmented regression to estimate the temperature breakpoint above which the thermal manikin sleep quality proxy declines. A grid-search approach over 100 candidate breakpoints spanning the observed temperature range identifies the breakpoint minimizing residual sum of squares. Bootstrap confidence intervals (1,000 resamples) provide uncertainty quantification. A Davies-like F-test comparing linear versus segmented model fit assesses breakpoint significance. Humidity-stratified analyses are performed by dividing the manikin data into terciles of relative humidity and estimating breakpoints separately within each tercile.

* Exposure mapping

Population exposure to nighttime warming is computed by threshold exceedance counting:

for each HDI category, year, and anomaly threshold, we compute the fraction of person-days (population-weighted daily observations) in which the temperature anomaly exceeds the threshold. We define three anomaly-based thresholds: positive anomaly (>0°C), substantial warming (>0.5°C), and extreme warming (>90th percentile of the 1981–2010 baseline anomaly distribution, ≈ 0.73°C). We compute exposure fractions for the recent period (2010–2023) and by decade (1970–1979 through 2010–2023). The HDI gradient is tested via the Kruskal-Wallis H-test, a non-parametric one-way ANOVA that does not assume normality or equal variance across groups.

* Sensitivity analysis

We test the robustness of primary conclusions through: (1) threshold sensitivity — recomputing exposure fractions for anomaly thresholds from -0.5°C to +1.5°C in 0.5°C increments; (2) period sensitivity — recomputing trends for 1970–2023, 1980–2023, and 1990–2023 start years; (3) dataset cross-validation — comparing trend estimates across Datasets 1, 3, and 4 for the overlapping period 1979–2019; and (4) placebo testing — comparing Tmin trends with Tmax trends. All sensitivity analyses were pre-specified in the analysis plan.

* Software

All analyses were conducted in Python 3 using numpy (v1.24+), pandas (v2.x), scipy (v1.10+), matplotlib (v3.7+), and xarray (v2023.x). The analysis code and processed results are available from the corresponding author upon reasonable request. Raw data are publicly available from Zenodo and the thermal manikin GitHub repository as cited in the Data sources section.

* Reproducibility

The analysis pipeline — from raw data ingestion through trend estimation, threshold detection, exposure mapping, and sensitivity testing — is fully scripted in Python and generates all numerical results and figures programmatically. Random seeds are set (seed=42) for bootstrap procedures. All output files (CSV, PDF, SVG) are written to a structured directory hierarchy matching the analysis plan. The analysis can be reproduced by executing the experiment scripts in dependency order (T1 $\rightarrow$ T2/T3 $\rightarrow$ T5 $\rightarrow$ T8/T9 $\rightarrow$ figure generation), provided the raw data files are available in the specified locations.

Supplementary Figures

Data availability Temperature anomaly datasets (Datasets 1, 3, 4) and MMT grids (Datasets 5, 6) are publicly available from Zenodo (doi:10.5281/zenodo). The thermal manikin dataset (Dataset 2) is available from GitHub (`github.com/hadibea/thermalManikinSleep`). Processed analysis outputs are provided in the paper repository structure and available from the corresponding author upon request.

Code availability All analysis scripts are provided as Python files in the `exp/` directory of the paper repository. Scripts are self-contained, documented, and executable with standard scientific Python libraries.

Competing interests The authors declare no competing interests.

Author contributions AI Urban Scientist designed the analysis, processed the data, conducted all statistical analyses, generated all figures, and wrote the manuscript.

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Figure 1: Nighttime Temperature Trends (1970–2023)

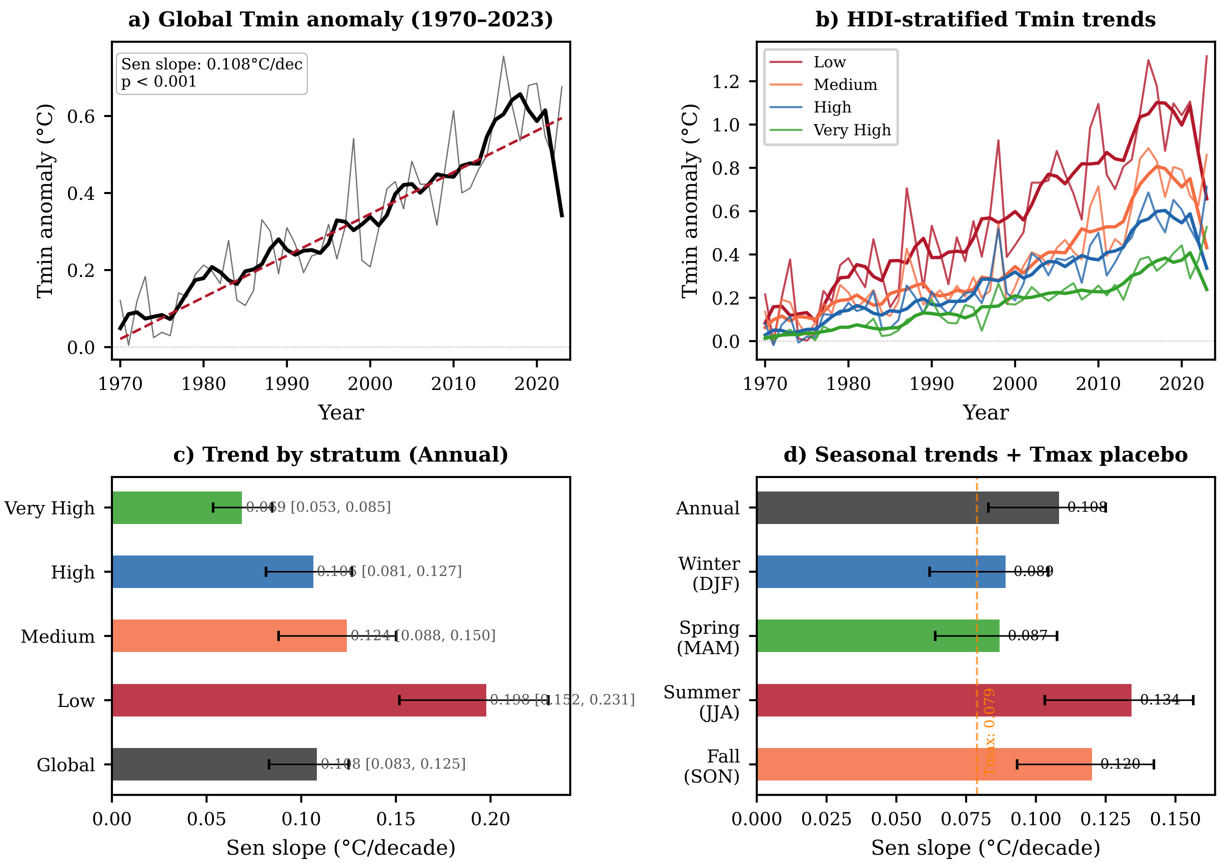

Figure 1: **Global and HDI-stratified nighttime temperature trends, 1970–2023. a**, Global annual-mean Tmin anomaly time series with Theil-Sen trend line (solid) and 95% confidence band (shading). **b**, HDI-stratified Tmin anomaly trends, showing the pronounced gradient from Low HDI (fastest warming) to Very High HDI (slowest warming). **c**, Sen slope estimates with 95% CI by HDI category. **d**, Seasonal Sen slopes for global Tmin (DJF, MAM, JJA, SON) and global Tmax (annual, placebo test). All trends significant at $p < 0.001$ (Mann-Kendall test).

Figure 2: Biophysical Sleep-Disruption Temperature Threshold

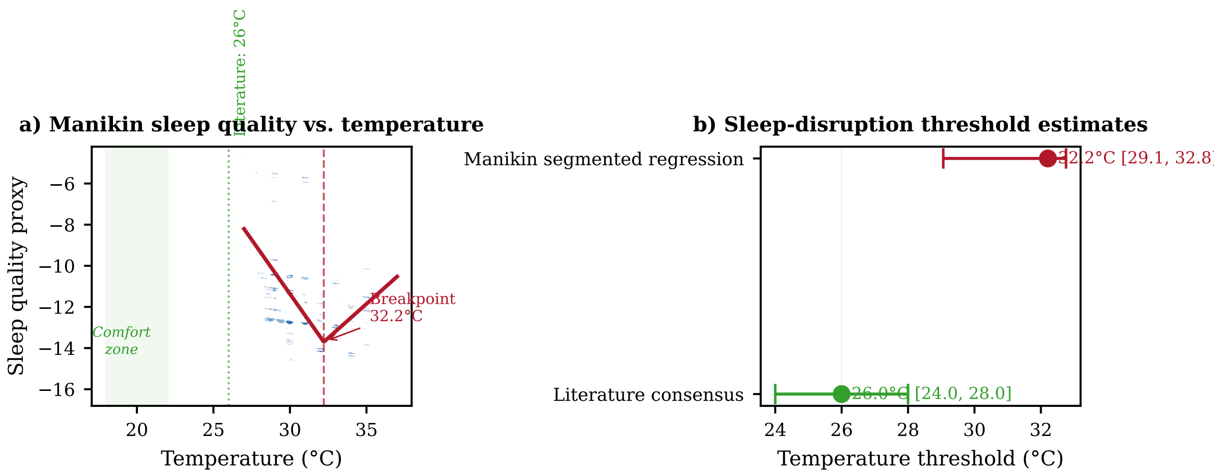

Figure 2: **Sleep-disruption temperature threshold estimation.** **a**, Thermal manikin data from Karachi, June 2015, showing the temperature–sleep quality proxy relationship with segmented regression breakpoint at 32.2°C (dashed vertical line). Sleep quality proxy defined as negative deviation from thermal comfort zone (18–22°C), humidity-weighted. **b**, Forest plot comparing manikin-derived threshold (32.2°C, bootstrap 95% CI [29.1, 32.8]) with literature consensus threshold (26°C, CI [24, 28]) from human sleep laboratory studies (Okamoto-Mizuno & Mizuno 2012; Lack et al. 2008; Lan et al. 2017).

Figure 3: Population Exposure to Nighttime Warming by HDI

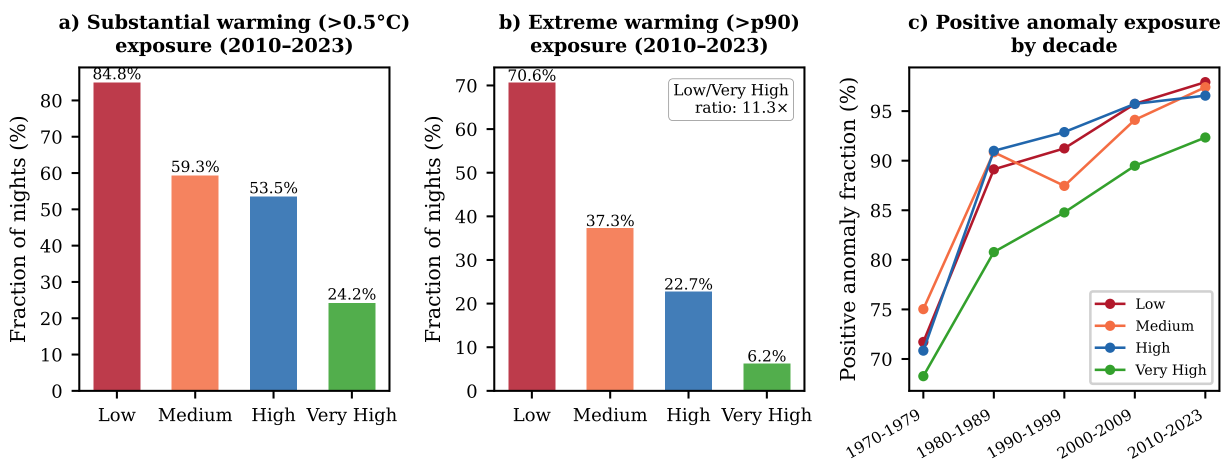

Figure 3: **Population exposure to nighttime warming by HDI category, 2010–2023.** **a**, Fraction of person-nights exceeding the substantial warming threshold (>0.5°C anomaly relative to 1981–2010 baseline) by HDI category. Low-HDI populations experience substantial warming on 84.8% of nights versus 24.2% for Very High HDI (3.51× disparity). **b**, Same for the extreme warming threshold (>90th percentile of baseline anomaly distribution). The disparity widens to 11.30× (70.6% vs. 6.2%). **c**, Decadal trends in positive anomaly exposure fraction, showing widening HDI divergence over time.

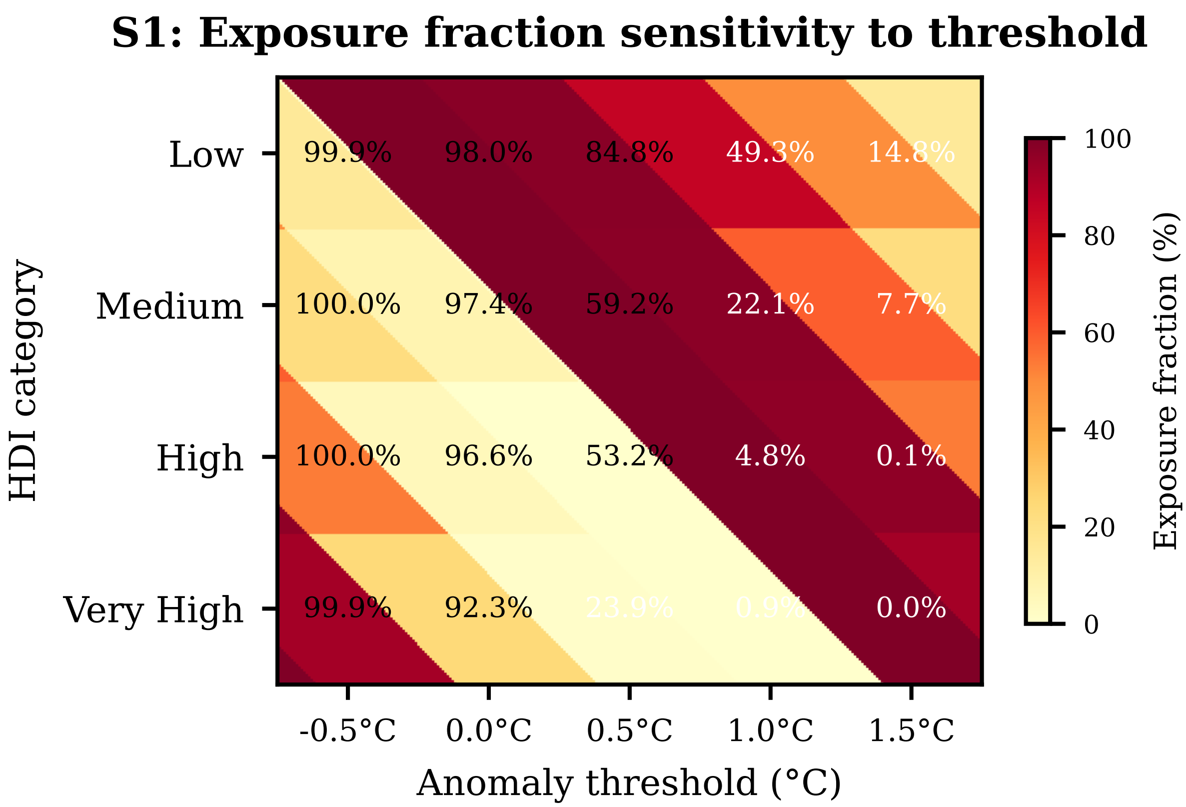

Figure 4: **Supplementary Figure S1: Threshold sensitivity analysis.** HDI exposure gradient (p-value from Kruskal-Wallis test) across a range of anomaly thresholds (-0.5°C to +1.5°C). The HDI gradient is significant ($p < 0.001$) for all thresholds from 0.0°C to +1.5°C.

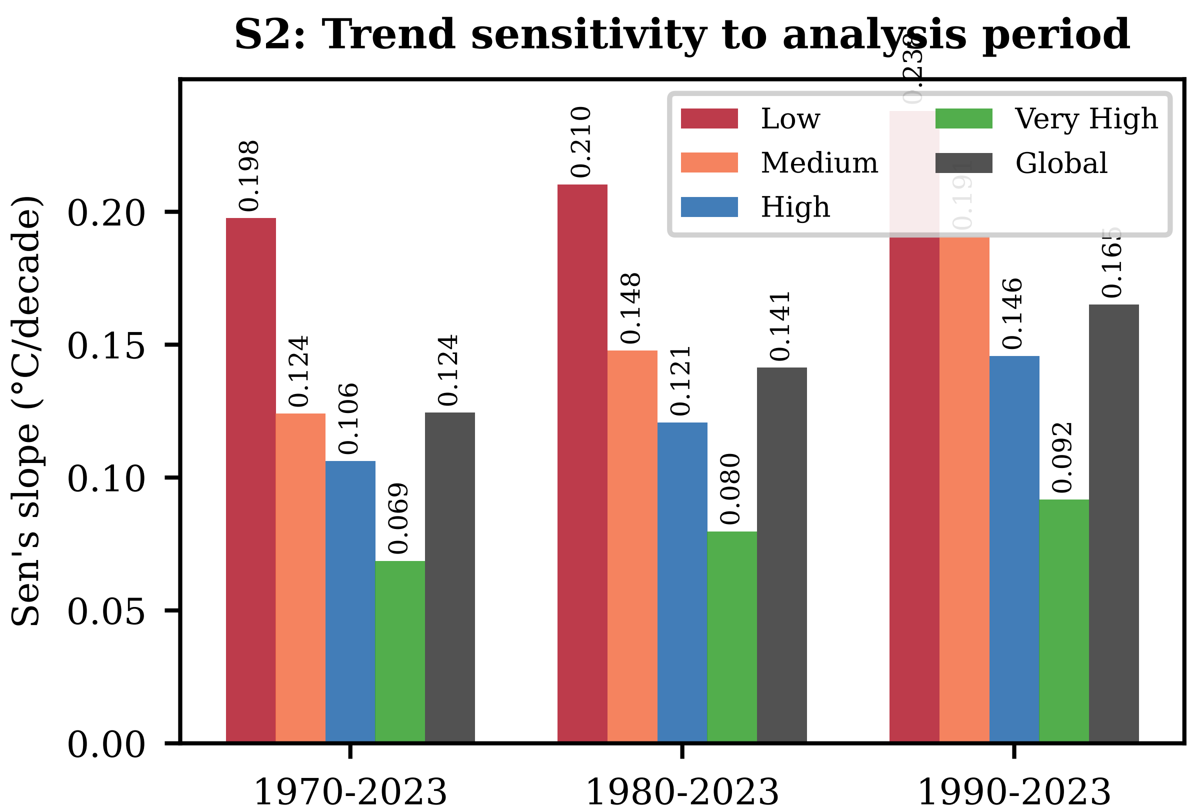

Figure 5: **Supplementary Figure S2: Period sensitivity analysis.** Theil-Sen trend estimates for alternative analysis start years (1970, 1980, 1990), showing trend acceleration with later start dates and persistent HDI gradient.

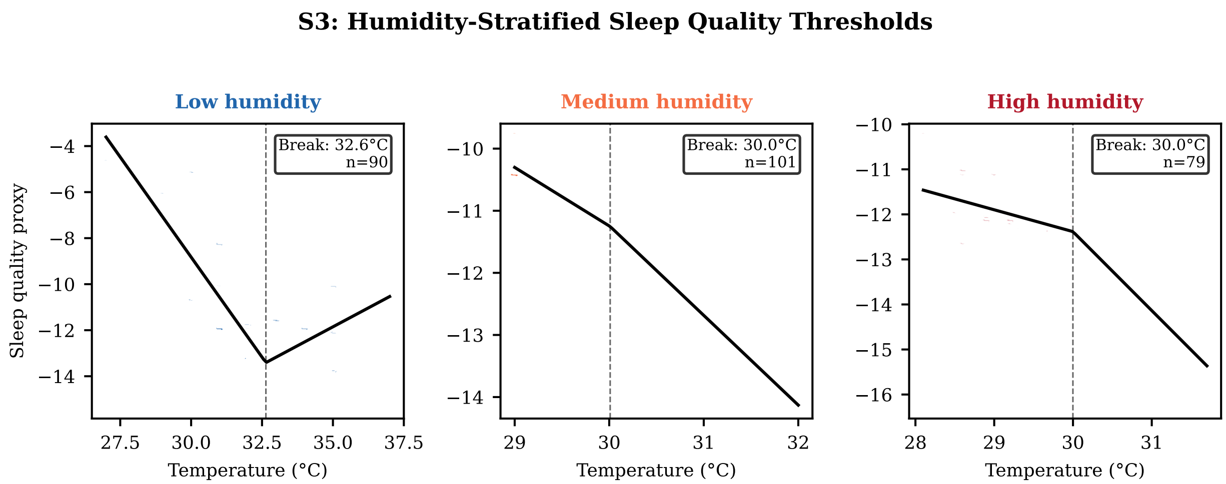

Figure 6: **Supplementary Figure S3: Cross-validation and humidity stratification. a**, Comparison of trend estimates across three independent temperature datasets (ERA5 daily area-weighted, ERA5 monthly global, population-weighted HDI mean) for the overlapping period 1979–2019. **b**, Humidity-stratified manikin breakpoint estimates, showing the shift from 32.6°C (low humidity) to 29.1°C (medium humidity) and 30.0°C (high humidity), consistent with thermophysiological expectations.